UNIT-II

Linear differential Equations

→ The standard form of higher order ordinary diff equation with const. coeff is of the form

$$\frac{d^n y}{dx^n} + k_1 \frac{d^{n-1} y}{dx^{n-1}} + k_2 \frac{d^{n-2} y}{dx^{n-2}} + \cdots + k_n y = X \quad \text{(I)}$$

where $k_1, k_2, k_3, \ldots, k_n$ are constant & X is a function in 'x'.

Linear D.E

A linear D.E is a diff equ in which dependent variable and its derivative occurs in only first degree but not multiplied together

Taking $D = \frac{d}{dx}$, $D^2 = \frac{d^2}{dx^2}$, $D^3 = \frac{d^3}{dx^3} \ldots$

Then equation (1) ⇒ $(D^n + k_1 D^{n-1} + k_2 D^{n-2} + \cdots + k_n)y = X$

Taking $F(D) \equiv D^n + k_1 (D)^{n-1} + \cdots + k_n$ (operator form of (1))

Then we have $\boxed{F(D)\, y = X}$ — (2)

→ First we find the complementary function (C.F)

IM

Def: Complementary function it is the solution of homogeneous linear differential equation.

That is the solution of $F(D)y = 0$

How to find?

Here we write the Auxiliary equation

i.e., Equating the operator $F(D)$ is to zero

⇒ $F(D) = 0$; $\boxed{D = m}$

Then the roots be $m_1, m_2, m_3, \ldots, m_n$

Depending on the nature of the roots, we write the complementary function

* **Case-①**: If the roots are real & distinct

then $\boxed{C.f = c_1 e^{m_1 x} + c_2 e^{m_2 x} + c_3 e^{m_3 x} + \cdots + c_n e^{m_n x}}$

* **case-(ii)** If any two roots are equal, $\boxed{m_1 = m_2}$

$$\boxed{C.f = (c_1 x + c_2) e^{m_1 x} + c_3 e^{m_3 x} + c_4 e^{m_4 x} + \cdots + c_n e^{m_n x}}$$

→ If any three roots are equal, $\boxed{m_1 = m_2 = m_3}$

$$\boxed{C.f = (c_1 x^2 + c_2 x + c_3) e^{m_1 x} + c_4 e^{m_4 x} + c_5 e^{m_5 x} + \cdots + c_n e^{m_n x}}$$

case-(iii) If any two roots are complex numbers

say $m_1 = \alpha + i\beta$ $\quad m_2 = \alpha - i\beta$

$$C.f = c_1 e^{(\alpha+i\beta)x} + c_2 e^{(\alpha-i\beta)x} = e^{\alpha x}[c_1(\cos\beta x + i\sin\beta x) + c_2(\cos\beta x - i\sin\beta x)]$$

$$\boxed{C.f = e^{\alpha x}[D_1 \cos\beta x + D_2 \sin\beta x]}$$

case-(iv) If any complex pair is repeated.

then $m_1 = m_2 = \alpha + i\beta$

$m_3 = m_4 = \alpha - i\beta$

$$C.f = (c_1 x + c_2) e^{(\alpha+i\beta)x} + (c_3 x + c_4) e^{(\alpha-i\beta)x}$$

$$\boxed{C.f = e^{\alpha x}[(D_1 x + D_2)\cos\beta x + (D_3 x + D_4)\sin\beta x]}$$

Problems on finding C.F

Pb $\frac{d^2y}{dx^2}+5\frac{dy}{dx}+6y=0$

Sol We write the operator form of the given equation (simbolic form)

$(D^2+5D+6)y=0$

Now Auxiliary equation is $f(D)=0$

i.e., $D^2+5D+6=0$

$\Rightarrow D=-2,-3$

$$\boxed{y=c_1e^{-2x}+c_2e^{-3x}}$$

Pb Solve diff equation $\frac{d^2x}{dt^2}+6\frac{dx}{dt}+9x=0$

$(D^2+6D+9)x=0$ $\because$ [Taking $D=\frac{d}{dt}$]

Now Auxiliary equation is $f(D)=0$

$D^2+6D+9=0$ 3 3

$D=-3,-3$

$$\boxed{x=(c_1t+c_2)e^{-3t}}$$

*

Pb $\frac{d^3y}{dx^3}+y=0$

The operator form is

$(D^3+1)y=0$

Now Auxiliary equation is $f(D)=0$

$D^3+1=0$

$D=-1$ is a root

By using synthatic method

$$\begin{array}{r|rrr} -1 & 1 & 0 & 0 \\ & 0 & -1 & -1 \\ \hline & 1 & -1 & 1 \;|\; 0 \end{array}$$

$D^2 - D + 1 = 0$

$$D = \frac{1 \pm \sqrt{-3}}{2} = \frac{1 \pm \sqrt{3}i}{2}$$

$\therefore D = -1, \frac{1+\sqrt{3}i}{2}, \frac{1-\sqrt{3}i}{2}$

$$\therefore \boxed{y = c_1 e^{-x} + e^{x/2}\left[c_2 \cos\frac{\sqrt{3}x}{2} + c_3 \sin\frac{\sqrt{3}x}{2}\right]}$$

(P6) $4y''' + 4y'' + y' = 0$

Given that

$$4\frac{d^3y}{dx^3} + 4\frac{d^2y}{dx^2} + \frac{dy}{dx} = 0$$

The opperator form is $(4D^3 + 4D^2 + D)y = 0$

Auxiliary form is $f(D) = 0$

$\cancel{4D^3 + 4D^2 + D = 0 \Rightarrow D[4D^2 + 4D + 1] = 0}$

$D = 0$ ~~is one root~~

By ~~using sythetic method~~

$4D^2 + 4D + D = 0$

$$\cancel{D = \frac{-4 \pm \sqrt{16-16}}{8} = \frac{-4}{8} = \frac{-1}{2}}$$

D

$$D\left[2D[2D+1] + [2D+1]\right] = 0$$

$D = 0 \quad D = \frac{1}{2} \quad D = \frac{1}{2}$

$y = c_1 + (c_2 + c_3)e^{-x/2}$

2M

(Pb) $(D^3+8)y=0$

Sol) Auxiliary equation is $D^3+8=0$

$\Rightarrow D=-2$ is root

$$\begin{array}{c|cccc} -2 & 1 & 0 & 0 & 8 \\ & 0 & -2 & 4 & -8 \\ \hline & 1 & -2 & 4 & 0 \end{array}$$

$D^2-2D+4=0$

$\Rightarrow D=\frac{2\pm\sqrt{4-16}}{2}$

$\boxed{D=1\pm i\sqrt{3}}$

$\therefore y=c_1e^{-2x}+e^{x}\left[c_2\cos\sqrt{3}x+c_3\sin\sqrt{3}x\right]$

(Pb) $\frac{d^4y}{dx^4}-y=0$

operator form $(D^4-1)y=0$

A.E in D^4-1

$(D^2)^2-(1)^2=0$

$(D^2-1)(D^2+1)=0$

$D=\pm 1, \pm i$

$\boxed{y=c_1e^{x}+c_2e^{-x}+c_3\cos x+c_4\sin x}$

Particular Integral

From given Non-Homo eq $F(D)\,y = x$, $y = \frac{1}{F(D)} x$ is called a particular integral for the given equation. Here $\frac{1}{F(D)}$ is the inverse operator

How to find particular integral

*case (i): Suppose $x = e^{ax}$

Then $P.I = \frac{1}{F(D)} e^{ax}$

we replace D by a

$= \frac{1}{F(a)} e^{ax}$ (provided $F(a) \neq 0$)

If $F(a) = 0$ then

$P.I = x \cdot \frac{1}{F'(D)} e^{ax}$

(replace D by a)

$= x \cdot \frac{1}{F'(a)} e^{ax}$ (provided $F'(a) \neq 0$)

If $F'(a) = 0$, then $P.I = x^2 \cdot \frac{1}{F''(D)} e^{ax}$

(replace D by a)

$= x^2 \cdot \frac{1}{F''(a)} e^{ax}$

Note:

(1) If RHS $= a^x$

$= e^{\log_e a^x}$

$= e^{x(\log a)}$

for ex: $(D^2 - 5D + 6)\,y = e^{4x} + 2^x$

Case-(ii): Suppose $X = \sin(ax+b)$ (or) $\cos(ax+b)$

Then $P.I = \frac{1}{F(D)} \sin(ax+b)$ (or) $\cos(ax+b)$

Here we replace $D^2 \longrightarrow -a^2$

$$\Rightarrow \frac{1}{F(D^2)} \sin ax = \frac{1}{F(-a^2)} \sin(ax+b)$$

(provided $F(-a^2) \neq 0$)

If $f(a^2) = 0$ then it is failure then lly to case (i)

Case-(iii) When $X = x^m$ (Polynomial in x)

then $P.I = \frac{1}{F(D)} x^m$

Here, we expand $[F(D)]^{-1} (x^m)$ and apply each term or the function x

Case-(iv) $X = e^{ax} \cdot V(x)$

Then $P.I = \frac{1}{F(D)} e^{ax} \cdot V(x)$

$$= \frac{1}{F(D+a)} e^{ax} \cdot V(x) \quad [\text{Relace } D \to D+a]$$

case-(v) [General method]

If X = Any other function

Here $P.I = \frac{1}{F(D)} X$

We use the formula,

$$\boxed{\frac{1}{D-a} X = e^{ax} \int X e^{-ax} \, dx}$$

Does the General / Complete solution of a given non Homogeneous linear diff equation with constant coefficient is

$$\boxed{\text{General sol} = C.F + P.I}$$

(Pb) Solve the diff equation

$$\frac{d^3y}{dx^3} - 6\frac{d^2y}{dx^2} + 11\frac{dy}{dx} - 6y = e^{-2x} + e^{-3x}$$

Sol Taking $D = \frac{d}{dx}$ the operator form of the equation

$$(D^3 - 6D^2 + 11D - 6)y = e^{-2x} + e^{-3x}$$

Now the Auxilary equation is

$$D^3 - 6D^2 + 11D - 6 = 0$$

$$\cancel{D^3 - 11D^2 + 6D = 1}$$

$$\cancel{D(D^2 - 11D + 6) = 1}$$

$$D = 1 \; \cancel{\text{is a root}}$$

So $D = 1$ is a root

& $D = 2, 3$ is two roots.

$\therefore y_c = C.F = C_1e^{x} + C_2e^{2x} + C_3e^{3x}$

$y_p = P.I$

$$= \frac{1}{D^3 - 6D^2 + 11D - 6}\left(e^{-2x} + e^{-3x}\right)$$

$$= \frac{1}{(D^3 - 6D^2 + 11D - 6)}e^{-2x} + \frac{1}{(D^3 - 6D^2 + 11D - 6)}\left(e^{-3x}\right)$$

$$= \frac{e^{-2x}}{-60} + \frac{e^{-3x}}{-120}$$

$$\therefore y = y_c + y_p = C_1e^{x} + C_2e^{2x} + C_3e^{3x} - \frac{e^{-2x}}{60} - \frac{e^{-3x}}{120}$$

1) $\frac{d^2y}{dx^2}+5\frac{dy}{dx}+6y=e^{-2x}\sin 2x$

Sol: Given equation in operator form

when $D=\frac{d}{dx}$

$$(D^2+5D+6)y=e^{-2x}\sin 2x$$

Now A.E is $D^2+5D+6=0$

$$\Rightarrow D=-2,-3$$

$$\therefore y_c=c_1e^{-2x}+c_2e^{-3x}$$

Now $y_p=\frac{1}{D^2+5D+6}(e^{-2x}\sin 2x)$

$$=e^{-2x}\frac{1}{(D-2)^2+5(D-2)+6}\sin 2x$$

$$=e^{-2x}\frac{1}{D^2+4-4D+5D-10+6}\sin 2x$$

$$=e^{-2x}\frac{1}{D^2+D}\sin 2x$$

$$=e^{-2x}\frac{1}{-2^2+D}\sin 2x$$

$$=e^{-2x}\frac{D+4}{D^2-16}\sin 2x \quad =e^{-2x}\frac{D+4}{-20}\sin 2x$$

$$=-\frac{e^{-2x}}{20}\left[2\cos 2x+4\sin 2x\right]$$

$$y=y_c+y_p$$

$$=c_1e^{-2x}+c_2e^{-3x}-\frac{e^{-2x}}{10}\left[\cos 2x+2\sin 2x\right]$$

(14) $\frac{d^4y}{dx^4} - y = \cos x \cdot \cosh x$

Given equation in operator form

when $D = \frac{d}{dx}$

$(D^4 - 1)y = \cos x \cosh x$

Now A.E in $D^4 - 1 = 0$

$(D^2-1)(D^2+1) = 0$

$D = \pm 1 \quad D = \pm i$

$\therefore y_c = c_1 e^{x} + c_2 e^{-x} + c_3 \cos x + c_4 \sin x$

Now $y_p = \frac{1}{D^4 - 1} \cos x \cdot \left(\frac{e^{x} + e^{-x}}{2}\right)$

$= \frac{1}{2}\left[\frac{1}{D^4 - 1} \cos x e^{x} + \frac{1}{D^4 - 1} e^{-x} \cos x\right]$

$= \frac{1}{2}\left[e^{x} \frac{1}{(D+1)^4 - 1} \cos x + e^{-x} \frac{1}{(D-1)^4 - 1} \cos x\right]$

$= \frac{1}{2}\left[e^{x} \frac{1}{D^4 + 4D^3 + 6D^2 + 4D} \cos x + e^{-x} \frac{1}{D^4 - 4D^3 + 6D^2 - 4D} \cos x\right]$

$= \frac{1}{2}\left[e^{x} \frac{1}{(-1^2)^2 + 4D(-1^2) + 6(-1^2) + 4D} \cos x + e^{-x} \frac{1}{(-1^2)^2 - 4D(-1^2) + 6(-1^2) - 4D} \cos x\right]$

$= \frac{-1}{5} \cos x \left[\frac{e^{x} + e^{-x}}{2}\right]$

$= \frac{-1}{5} \cos x \cosh x$

$y = y_c + y_p$

$y = c_1 e^{x} + c_2 e^{-x} + c_3 \cos x + c_4 \sin x - \frac{1}{5} \cos x \cosh x$

(10) $(D^3+2D^2+D)y = x^2e^{2x} + \sin^2 x$

(Pb) $(D^2-1)y = x\sin x + (1+x^2)e^x$

$$y_c = c_1e^x + c_2e^{-x}$$

$$y_p = \frac{1}{D^2-1} x\sin x + \frac{1}{D^2-1} e^x (x^2+1)$$

$$= \frac{1}{D^2-1} \text{I.P. of } xe^{ix} + \frac{1}{D^2-1} e^x(x^2+1)$$

$$= \text{I.P of } \left\{ \frac{1}{D^2-1} x e^{ix} \right\} + \frac{1}{D^2-1} e^x(x^2+1)$$

$$= \text{I.P of } \left\{ e^{ix} \frac{1}{(D+i)^2-1} x \right\} + e^x \frac{1}{(D+1)^2-1} (x^2+1)$$

$$= \text{I.P. of } \left\{ e^{ix} \frac{1}{D^2+2Di-2} x \right\} + e^x \frac{1}{D^2+2D} (x^2+1)$$

$$= \text{I.P of } \left\{ \frac{e^{ix}}{-2} \left[1 - \left(\frac{D^2+2Di}{2}\right)\right]^{-1} (x) \right\} + \frac{e^x}{2D} \left[\left(1+\frac{D}{2}\right)^{-1} (x^2+1)\right]$$

$$= \text{I.P of } \left\{ \frac{e^{ix}}{-2} \left(1 + \left(\frac{D^2+2Di}{2}\right) + \cdots\right) x \right\}$$

$$+ \frac{e^x}{2D} \left[\left(1 - \frac{D}{2} + \frac{D^2}{4} - \cdots\right)(1+x^2)\right]$$

$$= \text{I.P of } \left\{ \frac{e^{ix}}{-2} (x+0+i) \right\} + \frac{e^x}{2D} \left[1 + x^2 - \frac{1}{2}(2x) + \frac{1}{4}(2)\right]$$

$$= \text{I.P of } \left\{ -\frac{1}{2} (\cos x + i\sin x)(x+i) \right\} + \frac{e^x}{2} \int \left(1 + x^2 - x + \frac{1}{2}\right) dx$$

$$= \text{I.P of } \left\{ -\frac{1}{2} (x\cos x + i\cos x + i(x\sin x) - \sin x) \right\} + \frac{e^x}{2}\left(x + \frac{x^3}{3} - \frac{x^2}{2} + \frac{x}{2}\right)$$

$$= -\frac{1}{2}(\cos x + x\sin x) + \frac{xe^x}{2} + \frac{x^3e^x}{6} + \frac{x^2e^x}{4} + \frac{xe^x}{4}$$

Case (V)

(Pb) $(D^2+a^2)y = \tan ax$

A.E $D^2+a^2=0$ $\quad D = \pm ai$

$\therefore y_c = c_1\cos ax + c_2\sin ax$

Now $y_p = \frac{1}{D^2+a^2}(\tan ax)$

$$= \frac{1}{(D-ai)(D+ai)}(\tan ax)$$

$$= \left[\frac{A_1}{D-ai} + \frac{A_2}{D+ai}\right](\tan ax) = \frac{1}{2ai}\left[\frac{1}{D-ai} + \frac{1}{D+ai}\right]\tan$$

Now $\frac{1}{D-ai}\tan ax = e^{aix}\int \tan ax\, e^{-iax}\,dx$

$$\boxed{\frac{1}{D-a}x = e^{ax}\int x e^{-ax}\,dx}$$

$$= e^{iax}\int \frac{\sin ax}{\cos ax}(\cos ax - i\sin ax)\,dx$$

$$= e^{iax}\left[\int \sin ax\,dx - i\int\left(\frac{1-\cos^2 ax}{\cos ax}\right)dx\right]$$

$$= e^{iax}\left[\frac{-\cos ax}{a} - i\left[\frac{\log(\sec ax+\tan ax)}{a}\right] + \frac{i\sin ax}{a}\right]$$

$$= e^{iax}\left[\frac{-1}{a}\left(e^{-iax} + i\log(\sec ax+\tan ax)\right)\right]$$

replace i by $-i$

Here $\frac{1}{D+ai}\tan ax = e^{-iax}\left[\frac{-1}{a}\left(e^{iax} - i\log(\sec ax+\tan ax)\right)\right]$

$$y_p = \frac{1}{2ai}\left[\cancel{\frac{-1}{a}} + \frac{ie^{iax}}{a}\log(\sec ax + \tan ax) + \cancel{\frac{1}{a}} + \frac{ie^{-iax}}{a}\log(\sec ax + \tan ax)\right]$$

$$= \frac{1}{2ai}\left[\frac{-i}{a}\log(\sec ax + \tan ax)\left(\frac{e^{iax}+e^{-iax}}{2}\right)\right]$$

$$= \frac{-1}{a^2}\cos ax \log(\sec ax + \tan ax)$$

$$y = y_c + y_p$$

$$= c_1\cos ax + c_2\sin ax - \frac{1}{a^2}\cos ax\log(\sec ax + \tan ax)$$

(Pb)

$$\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = e^{e^x}$$

Operator form is $(D^2+3D+2)y = e^{e^x}$

A.E $D^2+3D+2 = 0$

$\therefore y_c = c_1e^{-x} + c_2e^{-2x}$

$$y_p = \frac{1}{D^2+3D+2}e^{e^x} = \frac{1}{(D+1)(D+2)}e^{e^x}$$

$$= \left[\frac{1}{D+1} - \frac{1}{D+2}\right]e^{e^x}$$

~~Here $\frac{1}{D+1}e^{e^x}$~~

put $e^x = t$

$e^x dx = dt$

$$= \frac{1}{D+1}e^{e^x} - \frac{1}{D+2}e^{e^x}$$

$$= e^{-x}\int e^{e^x}e^x dx + e^{-2x}\int e^{e^x}e^{2x}dx$$

$$= e^{-x}\left[\int e^t dt\right] + e^{-2x}\left[\int e^t\, t\, dt\right]$$

$$= e^{-x}\left[e^t\right] + e^{-2x}\left[e^t(t-1)\right]$$

$$= e^{-x}\left(e^{e^x}\right) + e^{-2x}\left(e^{e^x}\left(e^x-1\right)\right)$$

$$= e^{-x}\cancel{e^{e^x}} + e^{-x}\cancel{e^{e^x}} + e^{-2x}e^{e^x}$$

$$= e^{-2x}e^{e^x}$$

(Pb) $(D^2+4)y = \sec 2x$

Method of Variation of Parameters (finding P.I only)

Consider the 2nd order diff equation $\boxed{y'' + Py' + Qy = X}$

where P, Q & X are the function in x. It's complementary function is $\boxed{y_c = c_1 y_1 + c_2 y_2}$, where y_1, y_2 are the two solutions of $\boxed{y'' + Py' + Qy = 0}$.

Replacing c_1, c_2 by two new functions u, v such that

$$u = -\int \frac{X y_2}{W} dx$$

$$v = \int \frac{X y_1}{W} dx$$

where W = Wronskian

$$= \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1 y_2' - y_1' y_2 (\neq 0)$$

then, the particular integral by this method

$$\boxed{y_p = -y_1 \int \frac{X y_2}{W} dx + y_2 \int \frac{X y_1}{W} dx}$$

Note: Wronskian can be used to see wheather the sol are linearly independent (or) linearly dependent

i.e, If $W(y_1, y_2, \ldots, y_n) = 0$,

then $y_1, y_2, \ldots, y_n$ are linearly dependent

Otherwise they are linearly independent

(Pb) Solve the diff equation $(D^2+1)y = \operatorname{cosec} x$

Here, A.E is $D^2+1=0$

$D = \pm i$

$y_c = c_1 \cos x + c_2 \sin x$

Take $y_1 = \cos x$ $\quad y_2 = \sin x$

Now, Wronskian $w = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = 1 \; (\neq 0)$

By method of variation of parameters,

$$y_p = -y_1 \int \frac{X y_2}{W} dx + y_2 \int \frac{X y_1}{W} dx$$

$$= -\cos x \int \frac{\operatorname{cosec} x \sin x}{1} dx + \sin x \int \frac{\operatorname{cosec} x \cos x}{1} dx$$

$$= -x \cos x + \sin x \log(\sin x)$$

$y = y_c + y_p$

$$\boxed{y = c_1 \cos x + c_2 \sin x - x \cos x + \sin x \log(\sin x)}$$

(Pb) $(D^2+4)y = \tan 2x$

AE is $D^2+4=0$

$D = \pm 2i$

$y_c = c_1 \cos 2x + c_2 \sin 2x$

Take $y_1 = \cos 2x$ $\quad y_2 = \sin 2x$

$\therefore w = \begin{vmatrix} \cos 2x & \sin 2x \\ -2\sin 2x & 2\cos 2x \end{vmatrix} = 2 \; (\neq 0)$

By method of variation of parameters,

$$y_p = -y_1 \int \frac{X y_2}{W} dx + y_2 \int \frac{X y_1}{W} dx$$

$$= -\cos 2x \int \frac{\tan 2x \cdot \sin 2x}{2} + \sin 2x \int \frac{\tan 2x \cos 2x}{2} dx$$

$$= \frac{-\cos 2x}{2}\left[\int \frac{1-\cos^2 2x}{\cos 2x}\,dx\right] + \frac{\sin 2x}{2}\int \sin 2x\,dx$$

$$= \frac{-\cos 2x}{2}\left[\frac{\log(\sec 2x+\tan 2x)}{2}\right] + \cancel{\frac{\cos 2x}{2}}\,\cancel{\frac{\sin 2x}{2}} - \cancel{\frac{\sin 2x}{2}}\,\frac{\cos 2x}{2}$$

$$y_p = -\frac{1}{4}\cos 2x\left[\log(\sec 2x+\tan 2x)\right]$$

$$\boxed{y = y_c + y_p = c_1\cos 2x + c_2\sin 2x - \frac{1}{4}\cos 2x\left[\log(\sec 2x+\tan 2x)\right]}$$

(P.6) $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = e^{e^x}$

Sol $(D^2+3D+2)\,y = e^{e^x}$

A.E is $D^2+3D+2=0$

$D^2+2D+D+2=0$

$D(D+2)+1(D+2)=0$

$D=-1,-2$

$\boxed{y_c = c_1e^{-x} + c_2e^{-2x}}$

Here $y_1 = e^{-x}$ $y_2 = e^{-2x}$

$$W = \begin{vmatrix} e^{-x} & e^{-2x} \\ -e^{-x} & -2e^{-2x} \end{vmatrix} = -2e^{-3x} + e^{-3x} = -e^{-3x} \,(\neq 0)$$

$$y_p = -e^{-x}\int \frac{e^{e^x}e^{-2x}}{-e^{-3x}}dx + e^{-2x}\int \frac{e^{e^x}e^{-x}}{-e^{-3x}}dx$$

$$= e^{-x}\int e^x e^{e^x}dx - e^{-2x}\int e^{2x}e^{e^x}dx$$

$$\boxed{y_p = e^{-2x}e^{e^x}}$$

$y = y_c + y_p$

$$\boxed{y = c_1e^{-x} + c_2e^{-2x} + e^{-2x}e^{e^x}}$$

(b) $(D^2-2D+2)y = e^x \tan x$

A.E is $D^2-2D+2=0$

$D = 1 \pm i$

$$\boxed{y_c = e^x[C_1\cos x + C_2 \sin x]}$$

Take $y_1 = e^x \cos x$ $\quad y_2 = e^x \sin x$

$$W = \begin{vmatrix} e^x\cos x & e^x\sin x \\ -e^x\sin x + \cos x e^x & e^x\cos x + e^x\sin x \end{vmatrix}$$

$$= e^x\cos x[e^x\cos x + e^x\sin x] + e^x\sin x[e^x\sin x + \cos x e^x]$$

$$= e^{2x}\cos^2 x + \cancel{e^{2x}\sin x\cos x} + e^{2x}\sin^2 x - \cancel{e^{2x}\sin x\cos x}$$

$$= e^{2x} \quad (\neq 0)$$

By method of variation of parameters,

$$y_p = -y_1\int \frac{X y_2}{W}\,dx + y_2\int \frac{X y_1}{W}\,dx$$

$$= -e^x\cos x \int \frac{e^x\tan x \cdot e^x\sin x}{e^{2x}}\,dx + e^x\sin x\int \frac{e^x\tan x\, e^x\cos x}{e^{2x}}\,dx$$

$$= -e^x\cos x\int \frac{\sin^2 x}{\cos x}\,dx + e^x\sin x\int \sin x\,dx$$

$$= -e^x\cos x\log(\sec x + \tan x) + \cancel{e^x\sin x\cos x} + e^x\cos \ldots$$

$$\boxed{y_p = -e^x\cos x\log(\sec x + \tan x)}$$

$$\boxed{y = y_c + y_p = e^x[c_1\cos x + c_2\sin x] - e^x\cos x\log(\sec x + \tan x)}$$

(P3) $(D^2+1)y = \frac{1}{1+\sin x}$

A.E is $D^2+1=0$

$D = \pm i$

$$\boxed{y_c = c_1\cos x + c_2 \sin x}$$

Here $y_1 = \cos x$

$y_2 = \sin x$

y_p

$$W = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = 1 \ (\neq 0)$$

$$y_p = -\cos x \int \left(\frac{1}{1+\sin x}\right) \sin x \, dx + \sin x \int \frac{\cos x}{1+\sin x} dx$$

$$= -\cos x \left[\int \frac{1+\sin x}{1+\sin x} dx + \int \frac{1}{1+\sin x} dx\right] + \sin x \int \frac{\cos x}{1+\sin x} dx$$

$$= -\cos x \left[\int \left(1 - \frac{1}{1+\sin x}\right) dx\right] + \sin x \log(1+\sin x)$$

$$= -\cos x \left[x - \int \frac{1-\sin x}{1-\sin^2 x} dx\right] + \sin x \log(1+\sin x)$$

$$\boxed{y_p = -\cos x \left[x - \tan x + \sec x\right] + \sin x \log(1+\sin x)}$$

$y = y_c + y_p$

$$\boxed{y = c_1\cos x + c_2 \sin x - \cos x\left[x - \tan x + \sec x\right] + \sin x \log(1+\sin x)}$$

P6 $(D^2-2D+1)y = e^x \log(x)$

P6 $\cancel{(D^2-2D)}$

Sol A.E is $D^2-2D+1=0$

$D = 1, 1$

$$\boxed{y_c = c_1 e^x + c_2 x e^x}$$

Take $y_1 = e^x$ $\quad y_2 = x e^x$

$$w = \begin{vmatrix} e^x & xe^x \\ e^x & xe^x + e^x \end{vmatrix} = e^{2x}(1+x) - xe^{2x} = e^{2x}(1) \neq 0$$

$$y_p = -e^x \int \frac{e^x \log x (x) e^x}{e^{2x}} dx + xe^x \int \frac{e^x \log(x) e^x}{e^{2x}} dx$$

$$= -e^x \int x \log x \, dx + xe^x \int \log x \, dx$$

$$\cancel{= -e^x x(x\log x - x) + \frac{x^2}{2}}$$

$$y_p = -e^x \left[\log x \left(\frac{x^2}{2}\right) - \int \frac{1}{x} \times \frac{x^2}{2}\right] + xe^x(x\log x - x)$$

$$= -e^x \left[\frac{x^2 \log x}{2} - \frac{x^2}{4}\right] + x^2 e^x (\log x - 1)$$

$$= -e^x x^2 \left(\frac{\log x}{2} - \frac{1}{4}\right) + x^2 e^x (\log x - 1)$$

$$= e^x x^2 \left[(\log x - 1) - \left(\frac{\log x}{2} - \frac{1}{4}\right)\right]$$

$y = y_c + y_p$

$$\boxed{y = c_1 e^x + c_2 x e^x + e^x x^2 \left[(\log x - 1) - \left(\frac{\log x}{2} - \frac{1}{4}\right)\right]}$$

(Pb) $(D^2-2D)y = e^x \sin x$

Sol A.E is $D^2-2D=0$

$$D(D-2)=0$$

$$\boxed{D=0,2}$$

$$\boxed{y_c = c_1 + c_2 x e^{2x}}$$

Take $y_1 = 1$ $\quad y_2 = xe^{2x}$

$$W = \begin{vmatrix} 1 & xe^{2x} \\ 0 & 2xe^{2x}+e^{2x} \end{vmatrix} = e^{2x}(2x+1) \neq 0$$

$$y_p = -\int \frac{\cancel{e^x}\sin x \cdot x}{\cancel{e^{2x}}(2x+1)}\,dx + xe^{2x}\int \frac{\cancel{e^x}\sin x}{\cancel{e^{2x}}(2x+1)}\,dy$$

$$= -\int \frac{e^{x}\sin x \cdot x}{2x+1} + xe^{2x}\int \frac{\sin x}{e^{x}(2x+1)}\,dy.$$

Pb $(D^2-1)y = \dfrac{2}{1+e^x}$

Simultaneous System of equations

(16) Solve the simultaneous equation $\frac{dx}{dt} = 5x+y$, $\frac{dy}{dt} = y-4x$

Sol Given equations in operator-form

$Dx = 5x+y$ $\quad$ $Dy = y-4x$

(where $D = \frac{d}{dt}$)

$(D-5)x - y = 0$ — (1) $\quad$ $(D-1)y + 4x = 0$ — (2)

To eliminate 'y' from (1) & (2) operate $(D-1)\times(1) + (2)$

$(D-1)(D-5)x - \cancel{(D-1)y} = 0$

$4x + \cancel{(D-1)y} = 0$

$(D^2-6D+9)x = 0$ — (3)

eq (3) is a second order diff equation

AE is $D^2-6D+9=0$

$D = 3, 3$

$x = (c_1 t + c_2)e^{3t}$

Substituting 'x' in (1) we get

$(D-5)\left[(c_1 t + c_2)e^{3t}\right] = y$

$\Rightarrow y = c_1(3te^{3t} + e^{3t}) + c_2(3e^{3t}) - 5c_1 te^{3t} - 5c_2 e^{3t}$

$$\boxed{y = -2c_1 te^{3t} - 2c_2 e^{3t} + c_1 e^{3t}}$$

(10) Solve the simultanious equations $(D+2)x + 3y = 0$ —(1) ; $3x + (D-2)y = 2e^{2t}$ —(2)

To eliminate x we operate $3\times(1) - (D+2)(2)$

$$3\cancel{(D+2)x} + 9y$$

$$+\cancel{3(D+2)x} + (D+2)^2 y = (D+2)\,2e^{2t}$$

$$- \qquad + \qquad -$$

$$(9 + D^2 + 4 + 4D)\,y = 4e^{2t} + 4e^{2t}$$

$$(-D^2 - 4D + \boxed{13})\,y = 8e^{2t} \quad (3)$$

$$(D^2 + 4D - 5)\,y = -8e^{2t}$$

Eq (3) is a second order diff equation

A.E is $D^2 + 4D - 5 = 0$

$$\boxed{D = 1, -5}$$

$$\boxed{y_c = c_1 e^{-5t} + c_2 e^{t}}$$

$$y_p = \frac{1}{D^2+4D-5}(-8e^{2t})$$

$$= -8\,\frac{1}{4+8-5}(e^{2t})$$

$$= \frac{-8e^{2t}}{7}$$

$$\therefore \boxed{y = c_1 e^{-5t} + c_2 e^{t} - \frac{8e^{2t}}{7}}$$

(P6) solve $\frac{dx}{dt}+y=\sin t$; $\frac{dy}{dt}+x=\cos t$ given that $x=2$, $y=0$ at $t=0$

Sol Take $D=\frac{d}{dt}$

$$\therefore Dx+y=\sin t \quad \text{---} (1)$$

$$x+Dy=\cos t \quad \text{---} (2)$$

Eliminating 'y' we get

$$D^2x+Dy=D(\sin t)$$

$$x+Dy=\cos t$$

$$(D^2-1)x=0$$

$$\therefore \boxed{x=c_1e^t+c_2e^{-t}}$$

Substituting in (2) we get

$$c_1e^t+c_2e^{-t}+Dy=\cos t$$

$$\Rightarrow Dy=\cos t-c_1e^t-c_2e^{-t}$$

$$\Rightarrow y=\frac{1}{D}\left[\cos t-c_1e^t-c_2e^{-t}\right]$$

$$\boxed{y=\sin t-c_1e^t+c_2e^{-t}}$$

(P6) $(D+2)x-3y=5t$; $(D+2)y-3x=2e^{2t}$